Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/07/2005
PI Name:	Blancaflor, Elison B. Ph.D.		
Project Title:	Enhanced Labeling Techniques to Study the Cytoskeleton During Root Growth and Gravitropism		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	321-501-5368
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	Utilization and Life Science Office		
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City:	Kennedy Space Center	**State:**	FL
Zip Code:	32955	**Congressional District:**	8
Comments:	NOTE: Dr. Blancaflor moved to NASA Kennedy Space Center in spring 2021; as a PI, his prior grants were for work at the Noble Foundation in Oklahoma.		
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	09/01/2001	**End Date:**	03/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1518		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	An important function of the cytoskeleton lies in the regulation of plant form and orientation. Since gravity affects plant form, it is likely that the plant cytoskeleton is a target of gravity-induced modifications in plant morphogenesis. This proposal will use roots to study the dynamic behavior of the actin and microtubule cytoskeleton during gravity-induced differential growth in plants. Roots are ideal subjects for studies on the mechanisms of growth control because of the clear distinctions between developmental zones along the longitudinal axis (e.g. meristem, elongation and maturation zones) and because roots have a predictable growth response to gravity (i.e. curvature). The ability to observe these developmental stages in a single root and the predictable gravity-induced growth responses has allowed correlative studies between a specific type of cytoskeletal structure and the stage of cellular development. The proposed research will re-examine the actin cytoskeleton in gravistimulated roots using improved labeling methods that include green fluorescent protein (GFP)-cytoskeletal constructs for live cell imaging and an MBS crosslinking/glycerol permeabilization method for fixed roots. These approaches will be combined with classical hormone transport studies, computer based growth measurements and ion imaging using fluorescent sensors. Our experiments will be applied to roots of various species including Arabidopsis thaliana, Zea mays, Linum usitatissimum and Nicotiana benthamiana but will also employ Medicago truncatula because of the expanding genomic resources available at the Noble Foundation for this plant species. The objective of this proposal is to develop improved labeling methodology to help in our understanding of the function of the cytoskeleton during root growth and gravity responses. The results from this research will aid in developing a mechanistic knowledge on how gravity affects plant development at the cellular and molecular level. The cytoskeleton in particular is an attractive cellular system to investigate because of its importance in several fundamental cellular processes such as the the establishment of plant form and orientation. These traits are critical for the targeted improvement of plants not only to increase yields for food production but also to modify plant stature to reduce biomass in an advance life support system needed for extended space missions. Through this research, we have also provided unique educational opportunities to students at all levels. Educational and outreach programs at the Noble Foundation have provided bio-imaging resources for undergraduate/high school student interships and teacher training.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The objective of this research is to investigate the role of the cytoskeleton in root responses to gravity. Results from the proposed research will extend our understanding of how plants tell their correct orientation (i.e. know up from down). The ability of plants to respond to gravity is important for agriculture since crop productivity depends largely on plant form so it can more efficiently explore the soil for nutrients and water as well as optimize light absorption for photosynthesis. The cytoskeleton is critical to achieving the correct orientation of the plant thus the basic knowledge gained from this research should be helpful for optimizing strategies for crop production on Earth.
Task Progress:	This NASA funded project led to the publication of (10) peer reviewed papers with three of the ten papers featured on the journal cover. Our progress during the entire granting period is summarized below. We have shown that the actin cytoskeleton is likely to have a role in regulating plant gravitropism that is different from the predictions of current models. For instance, a popular explanation for actin's involvement in plant gravity sensing and signaling is that sedimenting amyloplasts (statoliths) pull on actin filaments that are linked to mechanosensitive ion channels in the plasma membrane. The activation of these channels then results in the influx of ions into the cytoplasm initiating a chain of events leading to organ bending. However, studies spanning several years indicate that the above model may need revision since plants with a disrupted actin cytoskeleton are still capable of initiating a gravitropic response (i.e. organs still bend upon horizontal reorientation). These observations are therefore contrary to models proposing that the actin cytoskeleton is involved in 'turning on' gravity signaling and response in plants. By conducting a detailed analysis of maize root gravitropism after altering the actin cytoskeleton with the potent actin disrupting drug Latrunculin B (Lat B), we have shown that both the sensitivity and response of roots to gravity is significantly increased. The enhanced gravity response was most pronounced when Lat B-treated roots were given a brief period of horizontal stimulation (ca. 10 min) followed by extended periods of rotation on a clinostat to randomize the gravity vector. Under such conditions, Lat B-treated roots would continue to curve strongly in the direction of the original gravity vector while control roots would bend weakly and in proportion to time of horizontal stimulation. These observations led us to propose that disrupting F-actin may be affecting the reset mechanisms that prevent the root from overshooting the vertical. Alternatively, these results could indicate that Lat B-treated roots remember the initial gravity stimulus despite randomizing the gravity signal with a clinostat. Although curvature of the root under these conditions is generally considered to be indicative of gravity sensing, this response could reflect the maintenance of the memory of the initial gravity vector. Without an intact and/or dynamic actin cytoskeleton, the initial gravity signal is locked, causing plants to overshoot the vertical. To probe further into the cellular mechanisms underlying the enhanced gravity response of roots induced by actin disruption, we conducted studies with seedling roots of Arabidopsis so we could exploit the molecular tools available in this model plant. We studied cellular events that are well known to be associated with root gravity responses including the development of asymmetric auxin gradients across opposite flanks of the root, cytoplasmic pH signaling in the gravity sensing columella cells of the root cap, and amyloplast dynamics and sedimentation in the columella cells. Our results show that all three of these events were modified by Lat B-treatment indicating that the actin cytoskeleton is involved in modulating these events. The amplification of the above signals when actin was disrupted with Lat B is consistent with actin's role as a down regulator of gravity signaling in roots. We have also made progress in developing improved methods for imaging the cytoskeleton in fixed and living plant tissue. We developed a set of novel in vivo F-actin reporters that are likely to further advance research on the plant actin cytoskeleton, particularly as it pertains to root development and gravitropism. These reporters are based on GFP fusions to Arabidopsis fimbrin 1 (AtFim1), an F-actin cross-linking protein that contains two actin binding domains (ABDs) arranged in tandem within a single polypeptide chain. We showed that GFP fusions to truncated variants of the AtFim1 protein, specifically ABD2, can decorate F-actin in living cells and allow the visualization of a finer and more dynamic F-actin network compared to the more popular Talin-GFP. More importantly, stable expression of these Fimbrin based GFP reporters in Arabidopsis allowed imaging of F-actin in different developmental regions of the root including the root cap, which is important for our studies on gravitropism. Also, we have generated tobacco BY2 cells and hairy roots of Medicago truncatula stably expressing these newly developed actin markers paving the way for detailed studies of actin turn over and dynamics in other important plant biological systems. Since we first published on this new F-actin reporter gene, we have had numerous requests for this construct from various researchers in the plant science community.

Bibliography Type:	Description: (Last Updated: 01/13/2026)
Articles in Peer-reviewed Journals	Wang, Y.-S., Motes, C.M., Mohamalawari, D.R., Blancaflor, E.B. "Green fluorescent protein fusions to Arabidopsis fimbrin 1 for spatio-temporal imaging of F-actin dynamics in roots" Cell Motility and the Cytoskeleton 59: 79-93 , Oct-2004
Articles in Peer-reviewed Journals	Hou, G., Kramer, V.L., Wang, Y.-S., Chen, R., Perbal, G., Gilroy, S., Blancaflor, E.B. "The promotion of gravitropism in Arabidopsis roots upon actin disruption is coupled with the extended alkalinization of the columella cytoplasm and a persistent lateral auxin gradient" Plant Journal 31: 113-125 , Jul-2004
Articles in Peer-reviewed Journals	Shin, H., Shin, H.-S., Guo, Z., Blancaflor, E.B., Masson, P.H., Chen, R. "Complex regulation of Arabidopsis AGR1/PIN2-mediated root gravitropic response and basipetal auxin transport by catharidin-sensitive protein phosphatases" Plant Journal 42: 188-200 , Apr-2005
Articles in Peer-reviewed Journals	Hou, G., Hill, J.P., Blancaflor, E.B. "Developmental anatomy and auxin response of lateral root formation in Ceratopteris richardii" Journal of Experimental Botany 397: 685-693 , Mar-2004
Presentation	Blancaflor, E.B. "Unraveling the ups and downs of plant gravity responses" Botany Department seminar series, Oklahoma State University, Stillwater, Oklahoma Mar-2004
Presentation	Blancaflor, E.B. "The actin cytoskeleton and its role in plant gravity responses" 15th Symposium in Plant Physiology, Pennsylvania State University, State College, Pennsylvania May-2004
Presentation	Blancaflor, E.B. "Unraveling the ups and downs of plant gravitropism" Botany Department seminar series, University of Oklahoma, Norman, Oklahoma Sep-2004
Presentation	Blancaflor, E.B. "Unraveling the ups and downs of plant gravitropism. Cytoskeletal and ion signaling in the root cap" of Environmental and Plant Biology seminar series, Ohio University, Athens, Ohio Oct-2004
Presentation	Blancaflor, E.B. "Unraveling the ups and downs of plant gravity response. " School of Biological Sciences seminar series, Washington State University, Pullman, Washington Nov-2004
Presentation	Blancaflor, E.B. "Unraveling the ups and downs of plant responses to gravity" Department of Biology seminar series, Oklahoma Christian University, Oklahoma City, Oklahoma Jan-2005